EVERY FINITE PARTITION OF A TOPOLOGICAL SPACE IS REFINED BY A FINITE STRATIFICATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The notions of a finite partition of a topological space and of a finite stratification (a finite partition into locally closed pieces), and the fact that the closure of a piece is closed. Set up in the author's section on partitions and stratifications, immediately before the counted statement. Fixes the object to be constructed and the pieces from which it is built.

Lemma 0.1. *Let X be a topological space. Let $X = \coprod X_i$ be a finite partition of X. Then there exists a finite stratification of X refining it.*

Proof. Let $T_i = \overline{X_i}$ and $\Delta_i = T_i \setminus X_i$. Let S be the set of all intersections of T_i and Δ_i. (For example $T_1 \cap T_2 \cap \Delta_4$ is an element of S.) Then $S = \{Z_s\}$ is a finite collection of closed subsets of X such that $Z_s \cap Z_{s'} \in S$ for all $s, s' \in S$. Define a partial ordering on S by inclusion. Then set $Y_s = Z_s \setminus \bigcup_{s'<s} Z_{s'}$ to get the desired stratification. □

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